SAGE X3 V12 IT SYSTEM ADMINISTRATION MANUAL

TECHNICAL GUIDE FOR ERPFULL & INTERNAL IT SUPPORT ENGINEERS

ERPFULL IT OPERATIONS

Doc Version: 1.0 (Official) | Scope: Pure IT Operations & Infrastructure

1.0 THE IT ADMINISTRATOR'S CHARTER & RESPONSIBILITY BOUNDARY

Vendor training frequently overwhelms internal IT engineers by dumping every single accounting, purchasing, and production procedure onto technical staff. **IT is NOT required to understand advanced financial accounting or industrial manufacturing recipes.**

The role of the IT Department in Sage X3 is strictly **Infrastructure Health, System Availability, Security Governance, User Provisioning, and Cloud Integration Pipelines**. IT builds and maintains the highway; the business departments (Finance, Supply Chain, Manufacturing) drive their own vehicles on it.

> ⚠️ **The Golden Rule for IT Sanity**
> If an incident involves server uptime, user account lockouts, network latency, deadlocks/hung screens, batch background crashes, print spoolers, or Cloud API pipelines, **it belongs to IT**.
> If an incident involves why debits do not equal credits, incorrect customer price list calculations, wrong raw material BOM ratios, or stock variances on physical shelves, **it strictly belongs to the Respective Business Department (HOD)**.

1.1 The Operational Boundary Matrix (What IT Owns vs. What Business Units Own):

OPERATIONAL DOMAIN	WHAT IT TRULY OWNS (IT SCOPE)	WHAT BUSINESS DEPARTMENTS OWN (NOT IT'S JOB!)
User & Access Control	• Creating user accounts & assigning security profiles. • Allocating license badges (`ERPFULL` , `ERPDIS` , `ERPMAN` , `ERPFIN`). • Resetting passwords & unlocking accounts.	• Approving business authorization levels. • Deciding monetary approval thresholds (e.g. PO limits).
System Uptime & Health	• Syracuse Web Tier (Node.js port 8124 / 443). • Batch Server (`Serveur de Batch`) uptime. • Clearing deadlocks & orphaned locks (`APLLCK`). • SQL Server backup maintenance & disk storage.	• Answering operational user questions on form inputs. • Resolving business workflow bottlenecks.
Printing & Document Outputs	• Sage Print Server service (`X3PRT`) connectivity. • Network printer mapping and driver stability.	• Modifying legal disclaimers or formatting on invoices. • Reconciling physical printouts against warehouse pallets.
Finance & Accounting	• Ensuring GL batch jobs run overnight without crashing. • Securing database financial tables and audit trails.	• **Reconciling Balance Sheets, P&L, or Trial Balances.** • Determining SST tax codes or depreciation rates.
Manufacturing & Logistics	• Setting up Site topology codes in Sage (`BOMSB` , `MY01`). • Ensuring API pools communicate with Cloud AIO.	• **Formulating BOM recipes, cycle times, or wastage rates.** • Conducting annual wall-to-wall stocktake counts.
Cloud Integration (BubblesO2 AIO)	• Maintaining Syracuse REST API pools (`/api1/x3/erp/...`). • Monitoring service account credentials (`WS_BUBBLESO2`).	• Deciding whether customer PO pricing is valid.

2.0 SAGE X3 ARCHITECTURAL TOPOLOGY FOR IT

Sage X3 V12 is built upon a decoupled 3-tier structure. Knowing which layer is failing solves 90% of technical tickets:

1. PRESENTATION TIER (CLIENT ACCESS) — `HTTPS / WSS`

Web Browsers (Chrome / Edge) on corporate workstations & tablet terminals | BubblesO2 AIO Cloud REST API Connector.

▼ **Port 443 / 8124 (SSL / TLS Reverse Proxy)**

2. SYRACUSE WEB ENGINE TIER — `Node.js · MongoDB · Elastic`

- **Node.js Cluster:** Runs Syracuse services on Port 8124. Handles authentication, routing, and REST APIs.
- **MongoDB (Port 27017):** Stores Syracuse users, license badges, screen layouts, and session states.
- **Elasticsearch (Port 9200):** Powers global search and quick navigation.

▼ **Port 50100 (AdxAdmin Internal Socket)**

3. X3 APPLICATION & 4GL RUNTIME TIER — `AdxAdmin · Batch · Print`

- **AdxAdmin (Port 50100):** Process agent managing user sessions.
- **Batch Server (Serveur de Batch):** Executes background automation (MRP, recurring allocations).
- **Print Server (X3PRT):** Spools Crystal Reports documents to network printers.

▼ **Port 1433 (TCP / SQL Server)**

4. DATABASE REPOSITORY TIER — `MS SQL Server (Instance: SAGEX3DATA)`

Stores all operational data schemas (Database: **x3v12**). Folders: **X3** (Mother), **BASE** (Production), **TESTX1** (Staging).

3.0 IT MASTER NAVIGATION CHEAT SHEET: TOP 15 FAST CODES

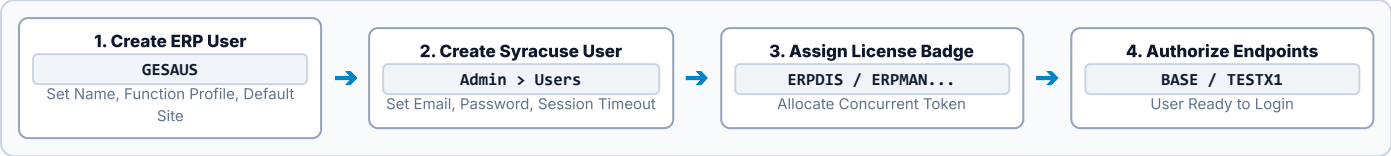
Never browse through deep menus. Type these fast codes directly in the top search bar:

FAST CODE	FULL FUNCTION NAME	NAVIGATION MENU PATH	WHY IT NEEDS THIS (PRIMARY PURPOSE)
GESAUS	User Master	*Administration > Users > Users*	Create and edit ERP user profiles, sites, and authorizations.
GESAUP	User Profiles	*Setup > Users > Function profiles*	Set access permissions by job role (e.g. Sales, Warehouse).
GESAFT	Functional Authorization	*Setup > Users > Functional authorization*	Restrict specific screens (e.g. block storekeepers from Finance).
ASYRREG	Batch Server Management	*Usage > Batch server > Server management*	**CRITICAL:** Monitor, pause, and restart the Batch background daemon.
UTILISAT	Active User Sessions	*Administration > Utilities > Sessions*	View logged-in users and disconnect hanging zombie sessions.
APLLCK	Locked Symbols Monitor	*Development > Utilities > Locks > Locked symbols*	**MOST USED:** Delete orphaned locks when a record gets stuck.
CONSSPR	Print Spool Queue	*Usage > Reports > Print queue*	Investigate stalled Crystal Reports jobs and printer errors.
CONSSST	Stock by Site Inquiry	*Stock > Inquiries > Stock by site*	Check physical stock vs. allocated vs. quarantine (Q) stock.
GESSMP	System Monitor	*Usage > System monitor*	Monitor server CPU, memory consumption, and active SQL queries.
DOSSIER	Folder Setup	*Development > Development setup > Folders*	Inspect folder versions, database schema names, and patch status.

4.0 USER ADMINISTRATION, SECURITY GOVERNANCE & LICENSE BADGES

4.1 Understanding the 2-Step User Identity Architecture

In Sage X3 V12, an active user requires two linked records: **Syracuse Web User** (manages credentials, badges, endpoints) and **ERP X3 User (GESAUS)** (manages default sites and approval limits):



4.2 Demystifying Sage X3 License Badges:

BADGE CODE	BADGE DISPLAY NAME	PERMITTED SYSTEM OPERATIONS	TARGET DEPARTMENTS
ERPFULL	Full ERP System	Unrestricted access across all modules: Finance, Distribution, Manufacturing, Admin tools, and Data Dictionary.	IT System Administrators, IT Leads, Operations Directors.
ERPDIS	Distribution	Sales Orders (`GESSOH`), Purchasing (`GESPOH`), GRN Receipts (`GESPTH`), Deliveries (`GESSDH`), and Stock.	Sales Executives, Warehouse Clerks, Storekeepers, Logistics.
ERPMAN	Manufacturing	Work Orders (`GESMFG`), BOM Recipes (`GESBOD`), Routing (`GESROU`), Tracking (`GESMTK`), and QC Release (`GESSCS`).	Production Planners, Plant Technicians, Shift Supervisors, QC.
ERPFIN	Financials	Customer Invoicing (`GESSIH`), Supplier Invoicing (`GESPIH`), Payments (`PAYMENT`), General Ledger, and Fixed Assets.	Finance Managers, Accounts Executives, Credit Controllers, Auditors.

4.3 How to Reset Passwords & Unlock Frozen Accounts:

Step-by-Step Account Unlock:
1. Navigate to **Administration > Users > Users**.
2. Search for the user's login ID.
3. If the user account shows *Status: Locked* (triggered after 3 failed password attempts), click **[Unlock]**.
4. To reset their password, click **[Edit]**, enter a temporary compliant password, and toggle on **"Force password change at next login"**. Click **[Save]**.

5.0 CLEARING SYSTEM LOCKS & ZOMBIE SESSIONS (APLLCK)

The #1 most frequent support call received by IT: *"Error: Record is locked by another user and cannot be edited!"*



FIELD NAME IN APLLCK	MEANING / DATA DISPLAYED	IT VERIFICATION ACTION
SYM (Symbol)	Table & Document Key (e.g. `SORDER~SOH2609001`).	Confirms which specific invoice, PO, or Work Order is locked.
USER	The login username holding the lock.	Check if user is actively typing or has walked away from their desk.
Date & Time	Timestamp when the lock was acquired.	If lock timestamp is hours old or from yesterday, it is a 100% dead lock.

6.0 BATCH SERVER (SERVEUR DE BATCH): SAFE MANAGEMENT (ASYRREG)

The **Batch Server** is the automated background engine of Sage X3. It processes night-run calculations, auto-allocations, and invoice runs. If it halts, users can still log in, but automated workflows silently freeze!

> ⚙️ **Monitoring & Restart Procedure (ASYRREG)**
> 1. Type **ASYRREG** in search bar (*Usage > Batch server > Server management*).
> 2. Check the status indicator: **Green = Running (Healthy)** | **Red = Stopped (CRITICAL!)**.
> 3. If Stopped: Click **[Deactivation]** in the Action Bar, wait 10 seconds, then click **[Activation]**.
> 4. Confirm status light turns Green and message states: *"Batch server running"*.

7.0 SYRACUSE WEB SERVICES & CLOUD INTEGRATION (BUBBLESO2 AIO)

The BubblesO2 AIO Cloud system communicates via Syracuse REST Web Services. IT must maintain this pipeline:

TECHNICAL PARAMETER	TARGET CONFIGURATION	IT OPERATIONAL SIGNIFICANCE
Protocol & Endpoint	`https://{HOST}:{PORT}/api1/x3/erp/{FOLDER}/{CLASS}`	Standard REST JSON format. Staging folder: `TESTX1`, Production: `BASE`.
Service Account	`WS_BUBBLESO2` (Basic Auth credentials)	Dedicated technical user. Never mix with personal staff accounts!
Badge Allocation	Must hold `ERPDIS` and `ERPFIN` (or `ERPFULL`)	Enables API to create Sales Orders, check Customer credit, and sync Invoices.
Session Timeout	Set to **0 (Infinite)** in Syracuse profile	Prevents Cloud queue workers from getting abruptly disconnected mid-sync.
Pool Recovery	*Admin > Web services > Classic SOAP pools*	If API returns *HTTP 503 Pool Exhausted*: Click **[Stop]**, wait 15s, then **[Start]**.

8.0 PRINT SERVER & CRYSTAL REPORTS INCIDENT RESPONSE

When warehouse or sales staff report: *"Print button hangs"* or *"Crystal Reports communication failed"*:

> **Quick Print Service Restart Steps:**
> 1. Log in to the Windows Server hosting Sage X3.
> 2. Open Windows Services: Press `Win + R`, type `services.msc`, and press Enter.
> 3. Scroll down to find service: **Sage X3 Print Server** (Service name: `AdxSrvImp` or `X3PRT`).
> 4. Right-click and select **Restart**. Spooled print jobs will immediately resume printing!

9.0 THE 5-MINUTE DAILY MORNING IT HEALTH CHECK

- [ ] **1. Syracuse Web Access:** Open browser, navigate to Syracuse login page. Confirm login screen loads instantly without delay.
- [ ] **2. Batch Server (ASYRREG):** Check status indicator is GREEN and active. Verify no failed recurring batch queries.
- [ ] **3. System Locks (APLLCK):** Inspect locked symbols. Clear any orphan locks older than 12 hours.
- [ ] **4. Server Storage Capacity:** Check Drive C: and Database Drive. Ensure minimum 20% free space is available.
- [ ] **5. Windows Services:** Verify *Syracuse*, *MongoDB*, *AdxAdmin*, and *Sage X3 Print Server* are in 'Running' state.

10.0 EMERGENCY IT INCIDENT RESPONSE CHEAT SHEET

REPORTED INCIDENT / ERROR	LIKELY ROOT CAUSE	IMMEDIATE IT RESOLUTION STEPS
"System Completely Down / 502 Bad Gateway"	Syracuse Node.js service crashed, or MongoDB service on Port 27017 stopped.	1. Remote into server. 2. Open `services.msc`, check *Syracuse* and *MongoDB*. 3. Start/Restart services in sequence: MongoDB first, then Syracuse.
"Record is locked by [USER] and cannot be modified"	User's browser crashed or network dropped while in edit mode, leaving an orphan symbol.	1. Open `APLLCK`. 2. Find symbol matching document ID. 3. Click `[Delete Line]` to release lock immediately.
"Nightly jobs did not run (MRP / Invoices missing)"	Batch Server (`Serveur de Batch`) halted or crashed.	1. Open `ASYRREG`. 2. Click `[Deactivation]`, wait 10s, then click `[Activation]`. 3. Verify batch queue resumes processing.
"Cannot Login: Max number of connections reached"	License badge limit exhausted due to abandoned inactive sessions left open.	1. Open `UTILISAT` (*Admin > Utilities > Sessions*). 2. Filter by idle time > 60 minutes. 3. Terminate ghost sessions to free up license badges.
"BubblesO2 Cloud Sync Fails: 401 Unauthorized"	Integration service account password expired or user account locked in Syracuse.	1. Open Syracuse *Admin > Users > Users*. 2. Find user `WS_BUBBLESO2`. 3. Unlock account and verify password credentials.
"Printing DO / Invoices hangs indefinitely"	Print Server service (`X3PRT` / `AdxSrvImp`) is frozen.	1. Open Windows `services.msc`. 2. Locate *Sage X3 Print Server* and click **Restart**.
"Accounting asks IT why Balance Sheet doesn't tally"	**NOT AN IT BUG!** Financial journal entries or GL posting error by Finance.	Politely inform Finance that IT administers system availability and data integrity; accounting ledger reconciliation is owned by Finance HOD.

IT Technical Escalation & Vendor Support Contacts

When an issue requires certified Sage vendor intervention (e.g. database corruption or core 4GL engine patch failure), log an official ticket with the Vendor Support Portal providing: (1) Exact Sage X3 Patch Release (`12.0.37 / 2025 R1`), (2) Syracuse Version (`12.22.5.4`), (3) Full Error Stack Trace from Syracuse Logs, and (4) Affected Folder (`BASE` / `TESTX1`).